

Agents Unite: A Git-Native Proof-of-Trust Ledger for Distributed Financial Intelligence

Parag Paul

RIG AI

parag@rigai.co

June–September 2025

Abstract

Autonomous LLM agents can synthesize market sentiment, news flow, and technical context at scale. Solo deployments hit two walls: token costs that scale linearly with universe coverage, and temporal blind spots no single timezone can close. We present **Agents Unite**, an open-source architecture that treats a single GitHub repository as a distributed financial memory layer. Each contributor spends about \$0.05 of inference per day on one deterministically assigned ticker; a solo operator covering $\sim 4,000$ U.S. names pays \$200–\$2,000 for the same day. Reports land as pull requests. Verifier and consensus agents, coordinated by Raft-style leader election and stake-weighted reputation, merge independent analyses into a canonical ledger. Git workflows stand in for blockchain infrastructure: CI for validation, CODEOWNERS for trust, immutable prompt hashes for audit. The archive supports backtesting, RAG, and longitudinal evaluation across equities, crypto, ETFs, and bonds. Readers who fork the ledger pay nothing. Per-operator token burden drops by four orders of magnitude. History becomes the moat. We propose an *Open Financial Memory Foundation*, Apache-style but scoped to verifiable market intelligence, to steward it.

1 Introduction

Financial markets generate narrative, sentiment, and price information faster than any individual, human or agent, can absorb. Retail and institutional participants increasingly delegate synthesis to LLM agents, yet three structural failures persist:

1. **Ephemeral memory.** Agent sessions discard work at termination; yesterday’s NVDA analysis must be regenerated at full cost.
2. **Budget exhaustion.** Covering $\sim 4,000$ U.S. tickers daily with multi-step agentic research costs on the order of hundreds to thousands of dollars in API tokens per run when priced at July 2026 list rates (Table 2) and a fixed report schema of $\sim 1,800$ input / ~ 700 output tokens plus harness overhead [3, 11, 30].
3. **Timezone asymmetry.** A cron job in Austin misses Tokyo open chatter, overnight Reddit spikes, and post-close filings.

Agents Unite [2] reframes the problem: instead of one operator researching the entire market, *the crowd researches one ticker each*. Assignment is deterministic (contributors cannot cherry-pick symbols), and outputs land in a versioned path `data/YYYY-MM-DD/TICKER/`. Git history becomes the asset: every belief is a commit, every merge is consensus, every fork is a derivative index.

This paper formalizes the architecture, draws analogies to distributed consensus and proof-of-stake, and outlines

governance via an Apache-style foundation. The biggest moat is not the code. It is history.

2 Related Work

Agents Unite sits at an intersection that no single prior line of work fully covers: verifiable multi-agent memory, crowdsourced financial forecasting, and Git-shaped data versioning.

Several projects pursue decentralized agent memory. OriginTrail anchors knowledge objects with on-chain Merkle proofs [31]; Akashik keeps provenance chains across peers without a central repository [8]; HadAgent ties agent serving to a proof-of-inference blockchain [24]. PolyLink and VeriLLM push in a related direction, making inference itself verifiable across heterogeneous nodes [38, 40]. None of them store a longitudinal record of *what* the agents concluded about specific assets over time. That gap matters. Folklore offers the closest empirical hint: when peers share resolved inference trees, recall@1 on BEIR corpora jumps sharply versus a single-node cache (Table 1) [18]. The implication for Agents Unite is that a warm, pooled ledger of answered ticker-day questions should dominate solo regeneration once the archive has depth.

Crowdsourced forecasting has a longer history. Estimote showed that open earnings estimates can beat sell-side consensus [13, 23], and later field experiments forced

Table 1: Federated inference-tree sharing vs. single-node semantic cache (recall@1, $\leq 2\%$ false-accept; Folklore / BEIR) [18].

Corpus	Single-node	Federated trees	Gain
SciFact	81.5%	98.2%	+20.5%
NFCorpus	77.3%	97.6%	+26.2%
FiQA	66.0%	94.5%	+43.3%

a redesign: visible peer estimates induced herding, so the platform went blind [10]. Static corpora like DLT-Sentiment-News [16] prove demand for open financial labels but ship no live contribution loop. Agents Unite differs on outputs (narrative reports, not scalar EPS), assignment (deterministic, not self-selected), and draft isolation. Section 3.3 returns to the herding question.

On the data side, lakeFS [36], Project Nessie [33], and DuckLake [12] bring Git-like branching to data lakes by splitting metadata from bulky objects. Agents Unite borrows the same split in reverse: small, reviewable reports stay in Git; analytics read from columnar snapshots (Section 6.1).

3 Problem Formulation

Let T denote the ticker universe ($|T| \approx 4,000$), C the daily active contributors, and c the per-report inference cost. Table 2 lists the July 2026 per-million-token rates used throughout. With Agents Unite’s fixed report schema ($\sim 1,800$ input and ~ 700 output tokens [3]), a single-ticker completion costs about \$0.001 on GPT-4o-mini [30] or DeepSeek V4-Flash [11]. Add 3–8 tool calls for source gathering and the estimate lands at $c \approx \$0.05$, with a range of \$0.02–\$0.08 depending on model and adapter.

Table 2: Recent published LLM API list prices (July 2026).

Model	Vendor page	Input \$/M	Output \$/M
GPT-4o-mini	OpenAI [30]	0.15	0.60
DeepSeek V4-Flash	DeepSeek [11]	0.14	0.28
Gemini 2.5 Flash-Lite	Google [20]	0.10	0.40

Solo coverage cost is $O(|T| \cdot c)$: one operator researching the full universe spends $\sim \$200$ – $\$2,000$ /day (Figure 1). Crowd coverage is $O(C \cdot c)$: each contributor spends only c , while readers fork the ledger at zero marginal inference cost. The economic gain is therefore **per-operator burden reduction**, not elimination of aggregate work:

- **Temporal diversity**: agents in London, Tokyo, and Austin capture disjoint information windows;
- **Model diversity**: Claude, GPT, Gemini, DeepSeek, and local Ollama instances decorrelate hallucination errors [5];
- **Focus diversity**: randomized slices (sentiment,

news, social, trading) turn assignment collisions into complementary reports.

3.1 Cost Model

Table 3 derives the per-report estimate from published per-million-token rates. Solo scenarios multiply by $|T|$; the crowd scenario multiplies by one assignment per contributor.

Table 3: Per-report cost model (July 2026 API rates).

Component	Tokens	Rate	Cost
Input (canonical prompt)	1,800	\$0.15/M [30]	\$0.00027
Output (structured report)	700	\$0.60/M [30]	\$0.00042
Harness overhead ($\sim 5\times$)	tool calls, browsing		\$0.0035
Typical Agents Unite report	budget to premium harness		\$0.05

AU read (\$0)

AU contrib (\$0.05)

Solo budget (\$200)

Solo mid (\$600)

Solo prem (\$2K)
→ USD/day

Figure 1: Average daily inference cost by coverage model for a $\sim 4,000$ -ticker universe. *AU contrib*: one Agents Unite contributor (\$0.05/day). *AU read*: fork and consume existing reports (\$0). *Solo budget/mid/prem*: single operator at \$0.05, \$0.15, and \$0.50 per ticker. Rates from Table 2 [11, 20, 30].

Table 4: Per-operator economics: solo vs. crowd (see Figure 1).

Model	Daily cost	Coverage
Solo (budget, \$0.05/ticker)	$\sim \$200$	$\sim 4,000$ tickers
Solo (mid-tier, \$0.15/ticker)	$\sim \$600$	$\sim 4,000$ tickers
Solo (premium, \$0.50/ticker)	$\sim \$2,000$	$\sim 4,000$ tickers
Agents Unite contributor	$\sim \\$0.05$	1 ticker
Agents Unite reader	\$0	all history

3.2 DIY Data vs. Contribute vs. Consume

Researchers choose among three paths:

1. **DIY (solo)**. One operator pays $O(|T| \cdot c)$ for full-universe research (Figure 1, Table 5), plus vendor inputs.
2. **Contribute**. Pay $\sim \$0.05/\text{day}$ for one assigned ticker and merge into the shared ledger. The contributor’s marginal cost buys coverage of *their* assignment while unlocking read access to everyone else’s history.
3. **Consume (fork/read)**. Pay $\$0$ inference to clone the archive and run backtests or RAG over accumulated `sentiment_score` series. Price bars for return alignment still require a market-data API; the avoided cost is regenerating years of narrative/sentiment research.

Agents Unite targets spend on **regenerated agent research** and **narrative alt-data**, not terminal price workflows (Table 5).

Table 5: Indicative annual stack costs (2025–2026 published or surveyed averages).

Stack element	Source / basis	\$/yr
Bloomberg Terminal (1 seat)	Industry-reported list [27]	2026 $\sim \$31,980$
Alt-data (avg. per dataset)	Neudata buyer survey [26]	$\sim \$80,000$
Alt-data (avg. fund budget)	Neudata ($n=60$ buyers) [26]	$\sim \$1.6\text{M}$
Retail EOD API (Tiingo Power)	Vendor pricing [35]	$\sim \$2,388$
Intraday/history API (Polygon Adv.)	Vendor pricing [32]	$\sim \$73,000$
Solo LLM full-universe (budget)	$\$200/\text{day} \times 365$ (Table 2)	$\sim \$73,000$
Solo LLM full-universe (mid)	$\$600/\text{day} \times 365$	$\sim \$219,000$
AU contributor	$\$0.05/\text{day} \times 365$	$\sim \$18$
AU yes/no verify (est.)	$\ll \$0.01/\text{check} \times 365$	$\ll \$4$
AU reader (inference)	Fork ledger	0

Cheap verification: yes/no and cross-encoders. Full research agents are expensive because they browse, tool-call, and write long reports. Verifiers need not repeat that work. A practical cost reduction is **claim checking**: take a peer’s posted report (or a single claim from it), retrieve the cited URL snippet, and ask a small model only for a binary judgment (“supported? yes/no”) with a short rationale. At GPT-4o-mini rates (Table 2), a 400-token input / 20-token output check is on the order of $\$10^{-4}$ per claim, orders of magnitude below regenerating the report. Where even that is too heavy, **cross-encoder** rerankers and NLI models score (claim, evidence) pairs jointly without open-ended generation [28,34], enabling local CPU verification of source support and duplicate detection before a generative verifier is invoked. Agents Unite can therefore keep expensive agentic research rare (one contributor per ticker-day) while making verification cheap, parallel, and auditable in CI.

Backtesting savings from multiple angles. Backtests usually pay twice: for prices (Table 6) and narrative features. A shared ledger removes the second bill: fork tagged releases and `load_reports.py` series instead of paying $365 \times |T| \times c$ to regenerate history or renewing alt-data feeds (Table 5). Recent reports remain searchable

for RAG, and optional attestations [38,40] can re-verify them without re-crawling sources.

Table 6: Recent published market-data API prices for backtesting labels (2026).

Product	Published rate	\$/yr
Tiingo Power (individual)	$\$30/\text{mo}$ [35]	360
Polygon Stocks Starter	$\$29/\text{mo}$ [32]	348
Polygon Stocks Advanced	$\$199/\text{mo}$ [32]	2,388
Bloomberg Terminal (1 seat)	$\$31,980/\text{yr}$ [27]	31,980

3.3 Precedent and Divergence from Estimimize

Estimimize showed that open crowds can improve earnings-expectation measurement when contributor counts are large [13,23]. Agents Unite shares the crowd premise but diverges on three axes. Outputs are structured narrative reports with sources, not scalar EPS/revenue points. Ticker assignment is deterministic, so contributors cannot concentrate on popular names. And drafts are independent until merge, approximating Estimimize’s post-2015 blind design [10]. The product is closer to a long-gestating belief census than to a competing IBES-style estimate feed.

Herding risk. Da and Huang [10] show that viewing peer forecasts causes contributors to underweight private information, improving individual accuracy while harming consensus. Agents Unite faces an analogous risk if early-merged reports, public leaderboards, or RAG over yesterday’s consensus become focal points that later agents imitate. Schema validation and MAD outlier rejection blunt crude spam, but they do not by themselves restore informational independence when agents share training corpora, news wires, or retrieved context.

Proposed mitigation. We propose weighting consensus not only by contributor count or stake, but by **source diversity**: reports whose cited URL domains, modality slices (sentiment/news/social/trading), and model families are more orthogonal receive higher influence in the weighted median. Homogeneous clusters (many accounts citing the same wire story through the same harness) should not dominate simply because n is large. Formal calibration of diversity weights, and whether diversity should enter before or after MAD filtering, is left to Phase 3–4 evaluation.

4 System Architecture

Figure 2 depicts the end-to-end pipeline. There is no central application server; **GitHub is the database**.

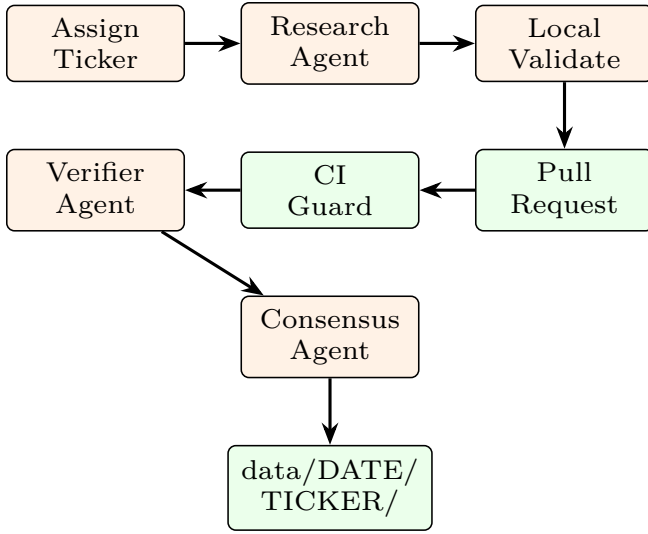


Figure 2: Distributed ingestion pipeline. Each contributor machine runs assignment, research, and validation locally; GitHub CI and peer agents provide trust anchors.

4.1 Deterministic Assignment

Ticker selection prevents manipulation:

$$k = \text{SHA256}(\text{date} : \text{contributor_hash}) \bmod |T| \quad (1)$$

The assigned ticker is T_k from a sorted, community-maintained universe file. Role selection uses a parallel hash lottery: $\sim 65\%$ research, $\sim 20\%$ verify, $\sim 15\%$ consensus [1].

4.2 Report Schema

Each artifact pair `report.*.md` + `sources.*.json` includes YAML frontmatter with `sentiment_score` $\in [-1, 1]$, `prompt_hash`, and `github_username`. Separating URLs into JSON keeps diffs readable and enables programmatic deduplication [3].

5 Consensus and Distributed Coordination

A single agent report is noisy. Agents Unite implements a **three-stage consensus pipeline** inspired by Raft [29] and proof-of-stake validator networks [15].

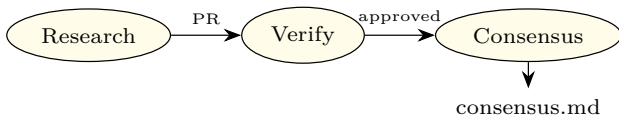


Figure 3: Role pipeline: research $\rightarrow$ verify $\rightarrow$ consensus.

5.1 Weighted Median Aggregation

Given n validated scores $\{x_i\}$:

- (i) Compute median m and MAD; reject outliers where $|x_i - m| > 2 \cdot \text{MAD}$.
- (ii) Compute weighted median with weights $w_i = \text{stake}_i \times \text{reputation}_i \times \text{recency}_i$.
- (iii) Assign confidence: *high* if $n \geq 3$ and spread ≤ 0.4 ; *low* if $n = 1$ [4].

Themes are *not* forced into agreement; divergence is preserved in `consensus.md`. Verifier roles should prefer the cheap yes/no and cross-encoder path in Section 3.2 before spending a full research budget to re-derive the same memo.

5.2 Raft Leader Election for Hourly Shards

Phase 3 introduces hourly updates. Multiple writers to the same `(date, ticker, hour)` shard would cause Git conflicts. Agents Unite adopts **deterministic Raft-style leader election** without a blockchain:

$$\text{leader} = \arg \min_i \text{SHA256}(\text{date} : \text{hour} : \text{ticker} : \text{hash}_i) \quad (2)$$

Among contributors who submitted valid hourly reports, the minimum hash wins write authority for the consensus merge, analogous to Raft’s single-leader principle [29] but requiring no network partition protocol because GitHub serializes merges.

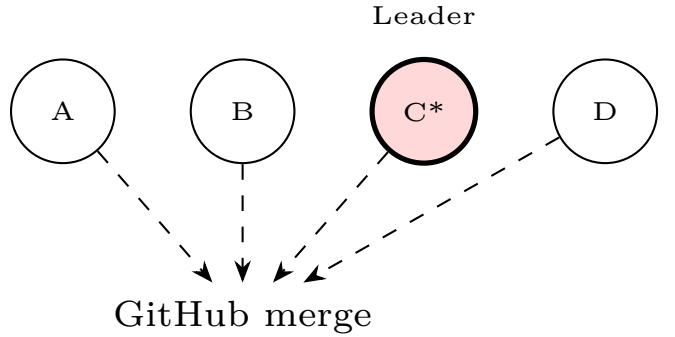


Figure 4: Hourly shard: contributor C wins deterministic leader election and merges `consensus.md`.

5.3 Proof-of-Stake Analogy

Ethereum’s proof-of-stake selects validators proportional to staked ETH [15]. Agents Unite’s Phase 4 maps this to **reputation staking**:

- **Stake** = accumulated accuracy vs. ex-post consensus and price outcomes;

- **Validators** = verifier agents with high merge approval rates;
- **Slashing** = reputation decay for rejected reports, fabricated URLs, or MAD outliers;
- **Rewards** = leaderboard visibility, weighted influence, eligibility for signal consumption.

No native token is required in Phase 1; reputation lives in `contributors/reputation.json`. Optional on-chain attestation layers (e.g., MINT Protocol [19]) can anchor Merkle roots of daily snapshots without storing private data on-chain.

6 Git as Consensus Substrate

Table 7 compares blockchain primitives to Git-native equivalents. Agents Unite occupies a middle ground: *permissioned trust* via GitHub identity and CI, with *permissionless read* via MIT license and fork.

Table 7: Blockchain $\leftrightarrow$ Git mapping in Agents Unite.

Blockchain	Agents Unite
Block	Commit
Validator set	Verifiers + CI
Consensus	Weighted median + merge
State trie	data/DATE/TICKER/
Smart contract	GitHub Actions workflows
Finality	Merged to main

6.1 Scalability and Operational Limits

Treating Git as a write-heavy database has a known failure mode. Package ecosystems that stored indexes in Git (crates.io’s Cargo index, Homebrew taps, and nixpkgs-scale trees) eventually hit clone, fetch, and delta-resolution costs that forced sparse HTTP indexes, CDN JSON feeds, or channel tarballs [25]. A financial memory ledger that accumulates thousands of daily report files faces the same asymptotics: directory width, packfile growth, and CI clone time degrade before cryptographic consensus becomes the bottleneck. Shallow clones and sparse checkouts delay pain; they do not remove the need for a read path that is not “git fetch the universe.”

Agents Unite therefore proposes a **hybrid storage path**: Git retains commits, PR review, CODEOWNERS, and per-report provenance (the collaboration substrate), while periodic **columnar snapshots** (Parquet/DuckDB-style exports keyed by date) serve analytics, RAG embedding jobs, and cold readers who need history without a full clone (mirroring the meta-data/data split in lakeFS, Nessie, and DuckLake) [12, 33, 36]. Phase 4 Merkle roots of those snapshots can be attested on-chain (e.g., MINT-style anchoring [19]) without moving private contributor data on-chain, connecting operational scale-out to the existing attestation

roadmap. In this design, Git remains the system of record for contribution and dispute; snapshots are the system of record for bulk consumption.

7 Foundation Governance

Sustainable open data requires neutral stewardship. We propose the **Open Financial Memory Foundation (OFMF)**, modeled on the Apache Software Foundation [6]:

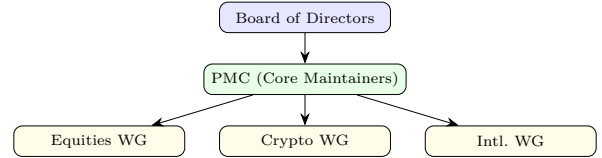


Figure 5: OFMF governance: Apache-style meritocracy with asset-class working groups.

Key principles: (i) individual merit over corporate affiliation; (ii) immutable core (`scripts/`, `agents/`) protected by CODEOWNERS; (iii) tagged dataset releases for reproducible backtests; (iv) 501(c)(3) structure to resist vendor capture.

8 Adversarial Robustness and the Contribution Attack Surface

Open ingestion expands the attack surface beyond spam. Three Phase 1 gaps remain; proposed defenses below are provisional.

Indirect prompt injection via ingested sources. Research agents retrieve filings, news pages, and social threads. Adversaries can embed instructions in those documents so that retrieved content overrides the agent’s intended task (indirect prompt injection) [21]. Schema checks and `prompt_hash` bind the *template*, not the retrieved payload. Verifier agents that re-read the same poisoned URLs may amplify rather than catch the attack. Candidate hardening directions include source sandboxing, explicit instruction/data delimiters, allowlisted domains, and verifier prompts that refuse to execute imperative language found in citations; none of these are asserted as sufficient without measurement.

Sybil and identity gaps in reputation. Phase 1 reputation lives in `contributors/reputation.json` keyed by GitHub identity. Cheap account creation, collusive rings, and recycled accuracy against a manipulable consensus remain open. MAD filters and merge-approval

rates raise the cost of naive spam but do not constitute proof-of-human or stake-slashing. Claims about long-run Sybil resistance should await Phase 4 attestation design and measured attack simulations against the reputation update rule.

Limits of CI as the only gate. Contributor Guard and offline schema validation block malformed PRs and protected-path edits; they do not judge factual correctness, source authenticity, or coordinated narrative capture. CI is a necessary integrity gate, not a sufficient epistemic one. Human review, verifier diversity, and expost accuracy scoring remain load-bearing, and currently under-specified relative to the threat.

9 Future Implications

For traders, one forkable ledger replaces repeated full-market agent runs. Consensus gating cuts single-model false positives. The cost arithmetic lands in Section 3.2: readers shift from DIY regeneration or vendor renewals toward contribute-or-fork paths, while price history stays à la carte.

For agent builders, canonical prompts and harness adapters create a portable reputation. A contributor who consistently merges accurate calls carries that record across employers and forks.

For society, the stakes are access. Narrative features locked behind terminal subscriptions become versioned and free to read. Global timezone coverage turns contributor geography into infrastructure.

The risks are real. Manipulation and poisoned sources require stronger attestation over time (Section 9). The system produces structured observations, not investment advice.

Regulatory context. FINRA [17] and SEC examination priorities [37] apply when firms use generative AI in advisory workflows. Agents Unite is an MIT-licensed observation archive, not a registered adviser; commercial wrappers that personalize consensus scores inherit those duties. “Observations, not advice” labeling remains a first-order control.

At scale (20,000 installs $\times$ 15% daily active $\approx$ 3,000 reports/day), Agents Unite approaches 75% daily universe coverage. No solo agent achieves that density at any price.

10 Agent Specialization and Analysis Modalities

Role randomization splits each day’s work into **complementary modalities**, so assignment collisions

become ensemble coverage rather than redundant spend. **Sentiment/social** agents synthesize narrative mood from forums and social feeds into a bounded **sentiment_score**, theme bullets, and attributed snippets. **Trading/full** agents ground those narratives in price, volume, range position, and catalyst tables. The **news** slice aggregates filings and headlines; verifiers cross-check cited numbers against source URLs, a lightweight peer analog to oracle checks. Figure 6 shows how randomized focus merges into daily **consensus.md**.

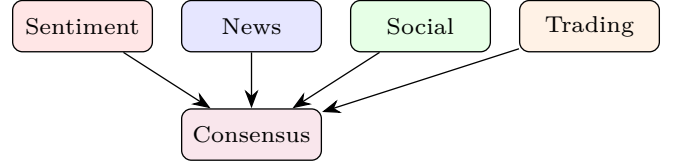


Figure 6: Complementary analysis modalities merge into daily **consensus.md**.

11 Asset Classes and Global Expansion

The U.S. 4,000-equity case is a baseline, not a boundary. For asset class a , solo annual inference is

$$K_a = N_a \times f_a \times c, \quad (3)$$

where N_a is the selected universe and f_a its annual research cadence. A contributor still pays for one assignment per cycle, while readers reuse the pooled history. Table 8 applies the same \$0.05/report assumption to representative scopes. Scenarios are independent, not additive; institutions should substitute their actual universe and cadence.

Table 8: Illustrative multi-asset inference economics. Universe counts use year-end 2025 / July 2026 sources; the bond row is an explicit watchlist scenario.

Universe	N_a	f_a/yr	Solo \$/yr
U.S. equities	4,000	365	73,000
Global listed companies [39]	60,889	365	1.11M
Global ETF products [14]	15,807	365	288,000
Regulated open-end funds [22]	148,692	52	387,000
Bond watchlist (illustrative)	100,000	52	260,000
Listed crypto assets [9]	8,165	365	149,000

The global equity count is a company-level proxy; share classes, preferreds, ADRs, REITs, and other listed instruments increase N_a when separately covered. Bonds require selection rather than exhaustive daily coverage: the market exceeds \$150 trillion outstanding [7], so issuer, maturity, liquidity, and event filters should define a weekly or event-driven watchlist. Crypto benefits from daily or hourly coverage, but only verified/liquid assets should enter the canonical universe. Under every scope,

Agents Unite changes the operator cost from $N_a f_{ac}$ to roughly f_{ac} per contributor (about \$18/year daily or \$2.60/year weekly), plus any external market-data fees; fork-only inference remains \$0.

Each class follows an Apache-style incubator: regional working groups prove schema fit, verifier coverage, and contributor density before joining the canonical history.

12 The Raft Census: Longitudinal Aggregation

We introduce the term **Raft census** to describe the periodic aggregation of independent agent observations into consensus snapshots over calendar time. Unlike a one-shot sentiment poll, the census is:

1. **Repeated:** daily (Phase 1), hourly (Phase 2+) observations per ticker;
2. **Independent:** contributors cannot see others’ drafts before submission;
3. **Reconciled:** verifiers and consensus agents produce `consensus.md`;
4. **Immutable:** raw reports persist alongside canonical merges in Git history.

This mirrors statistical sampling: each agent is an enumerator, verifiers are auditors, and consensus is official tabulation.

12.1 Backtesting Methodology

Quant researchers can construct sentiment factors from `consensus_score` or raw `sentiment_score` distributions:

- (a) Load reports via `examples/load_reports.py` for ticker t and window $[t_0, t_1]$;
- (b) Align scores to market close timestamps (configurable: UTC midnight or US close);
- (c) Compute forward returns r_{t+h} for horizons $h \in \{1, 5, 20\}$ days;
- (d) Test monotonicity: do high-sentiment quintiles outperform low-sentiment quintiles?
- (e) Attribute alpha to contributor subsets via `github_username` for reputation calibration.

Because every report includes `prompt_hash` and source URLs, backtests are **reproducible**, a requirement absent from proprietary terminal feeds.

12.2 RAG and LLM Downstream Queries

The Phase 2 pipeline embeds reports into a searchable index and knowledge graph (`wiki/`). Example queries enabled by longitudinal memory:

- “Which bearish Reddit themes preceded NVDA’s last twenty earnings misses?”
- “How did consensus sentiment on regional banks evolve during March 2023?”
- “Which contributors consistently led consensus by ≥ 3 days?”

These queries run against crowd-collected history at fork cost (Section 3.2).

12.3 Pending Empirical Evaluation

The following measurements are required before claiming production efficacy; `research/EXPERIMENTS.md` provides runbooks:

1. **Sentiment factor backtest:** quintile spreads and IC for $h \in \{1, 5, 20\}$; needs ≥ 60 trading days and a price API (Table 6).
2. **Ledger reuse vs. solo cache:** recall@1 at fixed false-accept on held-out ticker-day queries (Table 1).
3. **Verify cascade:** \$/decision and human agreement for re-research, yes/no, and cross-encoder NLI (Section 3.2).
4. **Source-diversity weights:** held-out calibration against equal-weight and stake-only aggregation (Section 3.3).
5. **Adversarial probes:** injection and Sybil attack-success rates after CI and verifier checks; then revise Section 9.

13 Comparison with Related Architectures

Table 9 positions Agents Unite against contemporary decentralized AI memory and inference systems.

Table 9: Agents Unite vs. related systems.

System	Git-native	Financial	Live
OriginTrail DKG	✗	✗	✓
Akashik	✗	✗	✓
HadAgent	✗	✗	✓
Folklore	✗	✗	✓
DLT-Sentiment	✗	✓	✗
Agents Unite	✓	✓	✓

Table 9 summarizes the trade-off: Git-native deployability over on-chain finality (Section 2), with optional Phase 4 anchoring for tamper evidence.

14 Conclusion

Agents Unite demonstrates that **blockchain ideas (consensus, stake-weighted trust, immutable history) need not require blockchain infrastructure**

when Git, CI, and open governance suffice. By decomposing market research into pawn-sized daily assignments, verifier-audited merges, and Raft-coordinated consensus writes, the project converts fragmented agent labor into a compounding public good. The moat is history; the mechanism is the crowd; the invitation is open.

Acknowledgments

The Agents Unite open-source community and the maintainers of github.com/rahiakil/agents-unite.

References

- [1] Agents Unite Contributors. Agent roles & pipeline. <https://github.com/rahiakil/agents-unite/blob/main/docs/ROLES.md>, 2026.
- [2] Agents Unite Contributors. Agents unite: Building the world’s financial memory. <https://github.com/rahiakil/agents-unite>, 2026.
- [3] Agents Unite Contributors. Architecture. <https://github.com/rahiakil/agents-unite/blob/main/docs/ARCHITECTURE.md>, 2026.
- [4] Agents Unite Contributors. Consensus design. <https://github.com/rahiakil/agents-unite/blob/main/docs/CONSENSUS.md>, 2026.
- [5] Agents Unite Contributors. Scientific methods. <https://github.com/rahiakil/agents-unite/blob/main/docs/METHODS.md>, 2026.
- [6] Apache Software Foundation. How the asf works. <https://apache.org/foundation/how-it-works/>, 2026.
- [7] Bank for International Settlements. International finance through the lens of bis statistics: Bond markets. https://www.bis.org/publ/qtrpdf/r_qt2509e.htm, 2025. Global debt securities outstanding exceeded \$150 trillion at year-end 2024.
- [8] Sahar Barak. Akashik: Decentralized cooperative memory for ai agents. <https://github.com/SaharBarak/akashik>, 2026.
- [9] CoinMarketCap. Cryptocurrency prices, charts and market capitalizations. <https://coinmarketcap.com/>, 2026. Approximately 8,165 verified/listed assets displayed in July 2026; automated token counts are much larger.
- [10] Zhi Da and Xing Huang. Harnessing the wisdom of crowds. *Management Science*, 66(5):1847–1867, 2020.
- [11] DeepSeek. Models and pricing. https://api-docs.deepseek.com/quick_start/pricing, 2026. V4-Flash: \$0.14/M input, \$0.28/M output tokens.
- [12] DuckDB Foundation. Ducklake: Sql as a lakehouse format. <https://ducklake.select/2025/05/27/ducklake-01/>, 2025.
- [13] Estimize. Crowdsourced earnings and economic estimates. <https://www.estimize.com/>, 2026.
- [14] ETFGI. Global etf industry, year-end 2025. etfgi.com, 2026. Reports 15,807 ETF products and 30,634 listings worldwide.
- [15] Ethereum Foundation. Proof-of-stake. <https://ethereum.org/en/developers/docs/consensus-mechanisms/pos/>, 2022.
- [16] ExponentialScience. Dlt-sentiment-news dataset. <https://huggingface.co/datasets/ExponentialScience/DLT-Sentiment-News>, 2025.
- [17] FINRA. Regulatory notice 24-09: Generative artificial intelligence in the securities industry. <https://www.finra.org/rules-guidance/notices/24-09>, 2024.
- [18] Folklore Contributors. Folklore: Federated inference-tree sharing for agent memory. github.com/usefolklore/folklore, 2026. BEIR recall@1 at $\leq 2\%$ false-accept: SciFact 81.5%→98.2%, NFCorpus 77.3%→97.6%, FiQA 66.0%→94.5% under federated tree-sharing vs. single-node semantic cache.
- [19] FoundryNet. Mint protocol: Verifiable proof-of-work for ai agents. <https://dev.to/foundrynet/what-is-mint-protocol-verifiable-proof-of-work-for-ai-agents-221f>, 2026.
- [20] Google. Gemini developer api pricing. <https://ai.google.dev/gemini-api/docs/pricing>, 2026. Gemini 2.5 Flash-Lite: \$0.10/M input, \$0.40/M output tokens.
- [21] Kai Greshake, Sahar Abdelnabi, Shailesh Mishra, Christoph Endres, Thorsten Holz, and Mario Fritz. Not what you’ve signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security*, 2023.
- [22] Investment Company Institute. 2026 investment company fact book. <https://www.ici.org/system/files/2026-04/2026-factbook.pdf>, 2026. Reports 148,692 regulated open-end funds worldwide at year-end 2025.

- [23] Russell Jame, Rick Johnston, Stanimir Markov, and Michael C. Wolfe. The value of crowdsourced earnings forecasts. *Journal of Accounting Research*, 54(4):1077–1110, 2016.
- [24] Landy Jimenez, Mariah Weatherspoon, Bingyu Shen, Yi Sheng, Jianming Liu, and Boyang Li. Hadagent: Harness-aware decentralized agentic AI serving with proof-of-inference blockchain consensus, 2026.
- [25] Andrew Nesbitt. Package managers keep using git as a database, it never works out. nesbitt.io, 2025. Dec. 2025; crates.io, Homebrew, and nixpkgs Git-index limits.
- [26] Neudata. The future of alternative data report. <https://www.neudata.co/blog/the-future-of-alternative-data-report-2024>, 2024. Survey of 60 buyers: avg. \$1.6M/year alt-data spend, ~\$80k per dataset, ~20 vendors.
- [27] NeuGroup. Bloomberg terminals: How much more you’ll pay next year. <https://connect.neugroup.com/en/public/blogs/bloomberg-terminals-how-much-more-youll-pay-next-year>, 2024. Reports 2025 renewal list prices of \$31,980/year (single seat) and \$28,320/year (multi-seat).
- [28] Rodrigo Nogueira and Kyunghyun Cho. Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085*, 2019.
- [29] Diego Ongaro and John Ousterhout. In search of an understandable consensus algorithm. In *Proceedings of the USENIX Annual Technical Conference*, 2014.
- [30] OpenAI. Api pricing. <https://openai.com/api/pricing/>, 2026. GPT-4o-mini: \$0.15/M input, \$0.60/M output tokens.
- [31] OriginTrail. Decentralized knowledge graph. <https://github.com/OriginTrail/dkg-v9>, 2026.
- [32] Polygon.io. Stocks api pricing. <https://polygon.io/pricing>, 2026. Stocks Advanced: \$199/month (\$2,388/year) for deep history and flat files.
- [33] Project Nessie. Transactional catalog for data lakes with git-like semantics. <https://projectnessie.org/>, 2026.
- [34] Hithesh Sankararaman, Mohammed Nasheed Yasin, Tanner Sorensen, Alessandro Di Bari, and Andreas Stolcke. Provenance: A light-weight fact-checker for retrieval augmented LLM generation output, 2024. Cross-encoder / NLI factuality scoring without LLM-as-judge.
- [35] Tiingo. Api pricing. <https://www.tiingo.com/about/pricing>, 2026. Individual Power plan: \$30/month (\$360/year).
- [36] Treeverse. lakefs: Data version control for data lakes. <https://docs.lakefs.io/>, 2026.
- [37] U.S. Securities and Exchange Commission. Examination priorities for fiscal year 2025. <https://www.sec.gov/files/2025-exam-priorities.pdf>, 2024.
- [38] Ke Wang, Zishuo Zhao, Xinyuan Song, Zelin Li, Libin Xia, Chris Tong, Bill Shi, Wenjie Qu, Eric Yang, and Lynn Ai. VeriLLM: A lightweight framework for publicly verifiable decentralized inference, 2026.
- [39] World Federation of Exchanges. Fy 2025 market highlights. world-exchanges.org/statistics, 2026. Reports 60,889 listed companies worldwide at year-end 2025.
- [40] Mingjin Zhang et al. Polylink: A blockchain based decentralized edge ai platform for llm inference. In *IEEE International Conference on Blockchain*, 2025.